COMPANION FOR CHAPTER 11

OF

FUNDAMENTALS OF PROBABILITY

WITH STOCHASTIC PROCESSES

FOURTH EDITION

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Chapter 11

Sums of Independent Random Variables and Limit Theorems

11A ADDITIONAL EXAMPLES

Example 11a Let X be a uniform random variable over the interval (a, b). For $n = 1, 2, \dots$, calculate $E(X^n)$.

Solution: $M_X(t)$, the moment generating function of X is given by

$$M_X(t) = \begin{cases} \dfrac{1}{b-a}\Big(\dfrac{e^{tb} - e^{ta}}{t}\Big) & \text{if } t \neq 0 \\[2ex] 1 & \text{if } t = 0. \end{cases}$$

(See Exercise 10, Section 11.1 of the book.) To find $E(X^n)$, we will use relation (11.2) of the book,

$$M_X(t) = \sum_{n=0}^{\infty} \frac{E(X^n)}{n!}\, t^n = 1 + \sum_{n=1}^{\infty} E(X^n) \cdot \frac{t^n}{n!}. \tag{1}$$

To do so, note that, for $t \neq 0$,

$$\begin{aligned} M_X(t) &= \frac{1}{t(b-a)}\Big[\sum_{n=0}^{\infty} \frac{(tb)^n}{n!} - \sum_{n=0}^{\infty} \frac{(ta)^n}{n!}\Big] \\ &= \frac{1}{t(b-a)}\Big[1 + \sum_{n=1}^{\infty} \frac{(tb)^n}{n!} - 1 - \sum_{n=1}^{\infty} \frac{(ta)^n}{n!}\Big] \\ &= \frac{1}{t(b-a)} \sum_{n=1}^{\infty} \frac{(tb)^n - (ta)^n}{n!} = \frac{1}{b-a} \sum_{n=1}^{\infty} \frac{b^n - a^n}{n!}\, t^{n-1} \end{aligned}$$

$$= 1 + \frac{1}{b-a}\sum_{n=2}^{\infty}\frac{b^n - a^n}{n!}\,t^{n-1} = 1 + \frac{1}{b-a}\sum_{n=1}^{\infty}\frac{b^{n+1} - a^{n+1}}{(n+1)!}\,t^n$$

$$= 1 + \sum_{n=1}^{\infty}\frac{b^{n+1} - a^{n+1}}{(b-a)(n+1)}\cdot\frac{t^n}{n!}.$$

Comparing this with relation (1), we have

$$E(X^n) = \frac{b^{n+1} - a^{n+1}}{(b-a)(n+1)}. \quad \blacklozenge$$

Example 11b The service times at a two-teller bank are independent exponential random variables each with parameter λ. At a random time Jim is being served by teller 2, Jeff is being served by teller 1, Judy and Julie are in the queue waiting to be served, and Judy is ahead of Julie. What is the probability that when Julie's service is completed Jim is still being served?

Solution: Let A_1 be the event that Jeff leaves the bank before Jim, A_2 be the event that Judy leaves the bank before Jim, and A_3 be the event that Julie leaves the bank before Jim. Using the memoryless property of exponential random variables, by symmetry, the desired probability is

$$P(A_1A_2A_3) = P(A_1)P(A_2 \mid A_1)P(A_3 \mid A_1A_2) = \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{8}.$$

An Alternative Solution: At the random time, let R be the remaining service time of Jim, let X_1 be the remaining service time of Jeff, and let X_2 and X_3 be the service times of Judy and Julie, respectively. Due the memoryless property of exponential random variables, R and X_1 are also exponential, each with parameter λ. The desired probability is $P(X_1 + X_2 + X_3 < R)$. Let $Z = X_1 + X_2 + X_3$; then Z is gamma with parameters 3 and λ. By (10.21),

$$\begin{aligned}
P(Z < R) &= \int_0^{\infty} P(Z < R \mid R = x)\lambda e^{-\lambda x}\,dx \\
&= \int_0^{\infty} P(Z < x)\lambda e^{-\lambda x}\,dx \\
&= \int_0^{\infty}\left[\int_0^x \frac{1}{2}\lambda e^{-\lambda t}(\lambda t)^2\,dt\right]\lambda e^{-\lambda x}\,dx \\
&= \int_0^{\infty}\left[\frac{1}{2}(\lambda^2 t^2 + 2\lambda t + 2)e^{-\lambda x}\right]_0^x \lambda e^{-\lambda x}\,dx \\
&= \int_0^{\infty}\Bigl(1 - e^{-\lambda x} - \lambda x e^{-\lambda x} - \frac{1}{2}\lambda^2 x^2 e^{-\lambda x}\Bigr)\lambda e^{-\lambda x}\,dx
\end{aligned}$$

$$= \left[-e^{-\lambda x} + \frac{1}{16\lambda^2} e^{-2\lambda x} \big(4\lambda^4 x^2 + 12\lambda^3 x + 14\lambda^2\right]_0^\infty$$

$$= 0 - \left[-1 + \frac{1}{16\lambda^2}(14\lambda^2)\right] = 1 - \frac{14}{16} = \frac{1}{8}. \quad \blacklozenge$$

Example 11c The service times at a two-teller bank are independent exponential random variables each with parameter λ. At a random time, Jim is being served by teller 2, Jeff is being served by teller 1, Judy and Julie are in the queue waiting to be served, and Judy is ahead of Julie. What is the probability that when Julie's service is completed Jim is still being served?

Solution: At the random time, let R be the remaining service time of Jim, let X_1 be the remaining service time of Jeff, and let X_2 and X_3 be the service times of Judy and Julie, respectively. Due the memoryless property of exponential random variables, R and X_1 are also exponential, each with parameter λ. The desired probability is $P(X_1 + X_2 + X_3 < R)$. Let $Z = X_1 + X_2 + X_3$; then Z is gamma with parameters 3 and λ. By relation (10.21) of the book,

$$\begin{aligned}
P(Z < R) &= \int_0^\infty P(Z < R \mid R = x)\lambda e^{-\lambda x}\,dx \\
&= \int_0^\infty P(Z < x)\lambda e^{-\lambda x}\,dx \\
&= \int_0^\infty \left[\int_0^x \frac{1}{2}\lambda e^{-\lambda t}(\lambda t)^2\,dt \right] \lambda e^{-\lambda x}\,dx \\
&= \int_0^\infty \left[\frac{1}{2}(\lambda^2 t^2 + 2\lambda t + 2)e^{-\lambda x} \right]_0^x \lambda e^{-\lambda x}\,dx \\
&= \int_0^\infty \Big(1 - e^{-\lambda x} - \lambda x e^{-\lambda x} - \frac{1}{2}\lambda^2 x^2 e^{-\lambda x}\Big)\lambda e^{-\lambda x}\,dx \\
&= \left[-e^{-\lambda x} + \frac{1}{16\lambda^2} e^{-2\lambda x} \big(4\lambda^4 x^2 + 12\lambda^3 x + 14\lambda^2\right]_0^\infty \\
&= 0 - \left[-1 + \frac{1}{16\lambda^2}(14\lambda^2)\right] = 1 - \frac{14}{16} = \frac{1}{8}.
\end{aligned}$$

An Alternative Solution: Let A_1 be the event that Jeff leaves the bank before Jim, A_2 be the event that Judy leaves the bank before Jim, and A_3 be the event that Julie leaves the bank before Jim. Using the memoryless property of exponential random variables, by symmetry, the desired probability is

$$P(A_1 A_2 A_3) = P(A_1)P(A_2 \mid A_1)P(A_3 \mid A_1 A_2) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}. \quad \blacklozenge$$

Example 11d A random variable X is called **chi-squared** with n degrees of freedom if its probability density function is given by

$$f(x) = \frac{x^{(n/2)-1}\, e^{-x/2}}{\Gamma(n/2)\, 2^{n/2}}, \qquad x > 0.$$

In Example 11.11 of the book, we showed that a chi-squared random variable with n degrees of freedom is gamma with parameters $(n/2, 1/2)$. By Exercise 22, Section 11.1 of the book, the moment generating function of a gamma random variable Y with parameters r and λ is

$$M_Y(t) = \Big(\frac{\lambda}{\lambda - t}\Big)^r, \qquad t < \lambda.$$

Therefore, the moment generating function of a chi-squared random variable X with n degrees of freedom is

$$M_X(t) = \Big(\frac{1/2}{1/2 - t}\Big)^{n/2} = (1-2t)^{-n/2}, \qquad t < 1/2.$$

Let $X_1, X_2, \ldots, X_k$ be independent chi-squared random variables with $n_1, n_2, \ldots, n_k$ degrees of freedom, respectively. Find the distribution function of $S = X_1 + X_2 + \cdots + X_k$.

Solution: By Theorem 11.3 of the book , for $t < 1/2$,

$$\begin{aligned} M_S(t) &= M_{X_1}(t) M_{X_2}(t) \cdots M_{X_k}(t) \\ &= (1-2t)^{-n_1/2}(1-2t)^{-n_2/2} \cdots (1-2t)^{-n_k/2} \\ &= (1-2t)^{-(n_1+n_2+\cdots+n_k)/2}. \end{aligned}$$

Since this is the moment generating function of a chi-squared random variable with $n_1 + n_2 + \cdots + n_k$ degrees of freedom, by Theorem 11.2 of the book, S is chi-squared with $n_1 + n_2 + \cdots + n_k$ degrees of freedom. We have shown that the sum of independent chi-squared random variables with $n_1, n_2, \ldots, n_k$ degrees of freedom is chi-squared with $n_1 + n_2 + \cdots + n_k$ degrees of freedom. ◆

Example 11e Suppose that, with equal probabilities, the value of a specific stock on any trading day increases 40% or decreases 30%, independently of the fluctuations of the stock value on the past and future trading days. Suppose that currently the stock price is $X_0 = 250$, and let X_n be the stock price after n trading days.

(i) Find the distribution of $\ln X_n$ for large n.

(ii) By finding X_n for large n, determine whether or not owning this specific stock over the long-term will turn out to be a mistake.

(iii) Find $\lim_{n\to\infty} E(X_n)$.

Solution: **(i)** In the first n trading days, let Y_n be the number of the days in which the price of the stock increases. Then Y_n is a binomial random variable with parameters n and $1/2$. Clearly,

$$X_n = 250(1.4)^{Y_n}(0.7)^{n-Y_n}.$$

This gives

$$\begin{aligned}\ln X_n &= \ln 250 + Y_n \ln 1.4 + (n - Y_n)\ln 0.7\\ &= \ln 250 + Y_n \ln\frac{1.4}{0.7} + n\ln 0.7\\ &= \ln 250 + Y_n \ln 2 + n \ln 0.7.\end{aligned}$$

Now, as $n \to \infty$, by the central limit theorem, Y_n is approximately $N\big(n/2, n/4\big)$. (See self-test problem 8, Chapter 11 of the book.) So $\ln X_n$ is also approximately normal with mean

$$E(\ln X_n) = \ln 250 + \frac{n}{2}\ln 2 + n\ln 0.7 \approx 5.52 - (0.0101)n,$$

and variance

$$\text{Var}(\ln X_n) = (\ln 2)^2 \text{Var}(Y_n) = (\ln 2)^2 \cdot \frac{n}{4} \approx (0.12)n.$$

(ii) Note that

$$X_n = 250\Big(\frac{1.4}{0.7}\Big)^{Y_n}(0.7)^n = 250\Big(2^{Y_n/n}\Big)^n(0.7)^n.$$

By the strong law of large numbers, as $n \to \infty$, $Y_n/n \to 1/2$ and $(0.7)^n \to 0$. Therefore, for large n, $X_n \to 0$ and owning this specific stock over the long-term will turn out to be a disaster. $\big[$For large n, $X_n \to 250(2^{1/2} \cdot 0.7)^n \approx 250(0.9899)^n$, which goes to 0 as n goes to infinity.$\big]$

(iii) Note that

$$E(X_{n+1}|X_n) = (1.4X_n)\cdot\frac{1}{2} + (0.7X_n)\cdot\frac{1}{2} = (1.05)X_n.$$

Thus

$$E(X_{n+1}) = E\big[E(X_{n+1}|X_n)\big] = (1.05)E(X_n).$$

This implies that

$$\begin{aligned}E(X_1) &= (1.05)E(X_0) = (1.05)(250)\\ E(X_2) &= (1.05)E(X_1) = (1.05)^2(250)\\ E(X_3) &= (1.05)E(X_2) = (1.05)^3(250)\\ &\vdots\\ E(X_n) &= (1.05)E(X_{n-1}) = (1.05)^n(250)\end{aligned}$$

So, as $n \to \infty$, $E(X_n) \to \infty$. ♦

Example 11f For $n \geq 1$, let X_n be a gamma random variable with parameters n and λ. Let F_n be the distribution function of $Y_n = X_n/n$. Let F be the distribution function of the constant random variable $X = 1/\lambda$. Show that, for all t, $t \neq 1/\lambda$, $\lim_{n\to\infty} F_n(x) = F(x)$.

Solution: Since

$$M_X(t) = E(e^{tX}) = E(e^{t/\lambda}) = e^{t/\lambda},$$

by the Lévy Continuity Theorem, it suffices to show that

$$\lim_{n\to\infty} M_{Y_n}(t) = e^{t/\lambda}.$$

Note that since

$$F(t) = \begin{cases} 0 & t < 1/\lambda \\ 1 & t \geq 1/\lambda \end{cases}$$

is continuous at all points except $t = 1/\lambda$, the relation $\lim_{n\to\infty} M_{Y_n}(t) = e^{t/\lambda}$ implies that, for all t, $t \neq 1/\lambda$, $\lim_{n\to\infty} F_n(x) = F(x)$. We will now establish the convergence of $M_{Y_n}(t)$ to $e^{t/\lambda}$. Noting that

$$M_{X_n}(t) = \Big(\frac{\lambda}{\lambda - t}\Big)^n, \qquad t < \lambda,$$

it follows that

$$\begin{aligned} M_{Y_n}(t) &= E(e^{tY_n}) = E\big(e^{tX_n/n}\big) = M_{X_n}(t/n) \\ &= \Big(\frac{\lambda}{\lambda - t/n}\Big)^n = \Big(\frac{n\lambda}{n\lambda - t}\Big)^n, \qquad t < n\lambda. \end{aligned}$$

Now

$$\begin{aligned} \lim_{n\to\infty} M_{Y_n}(t) &= \lim_{n\to\infty} \Big(\frac{n\lambda}{n\lambda - t}\Big)^n = \lim_{n\to\infty} \Big(\frac{n\lambda - t}{n\lambda}\Big)^{-n} \\ &= \lim_{n\to\infty} \Big(1 - \frac{t/\lambda}{n}\Big)^{-n} = \lim_{n\to\infty} \frac{1}{\Big(1 - \dfrac{t/\lambda}{n}\Big)^n} = \frac{1}{e^{-t/\lambda}} = e^{t/\lambda}, \end{aligned}$$

where the penultimate equality follows from the fact that $\lim\limits_{n\to\infty} \Big(1 + \dfrac{x}{n}\Big)^n = e^x$. ♦

Example 11g Convergence in Distribution Let X_1, X_2, ... be a sequence of random variables with distribution functions F_1, F_2, ..., respectively. Let X be a random variable with distribution function F. If

$$\lim_{n\to\infty} F_n(x) = F(x)$$

at all points in which F is continuous, then we say that X_n **converges in distribution** to the random variable X. Note that, with this definition, the Lévy Continuity Theorem states that, for a sequence of random variables $X_1, X_2, \ldots$, if $M_{X_n}(t)$ converges to $M_X(t)$, the moment generating function of a random variable X, then X_n converges to X in distribution.

Let $U_1, U_2, \ldots$ be a sequence of independently selected random numbers from the interval $(0, 1)$. For $n \geq 1$, let $X_n = \min(U_1, U_2, \ldots, U_n)$. Let X be the constant random variable $X = 0$. Show that X_n converges to X in distribution.

Solution: The distribution function of X is given by

$$F(x) = \begin{cases} 0 & x < 0 \\ \\ 1 & x \geq 0 \end{cases}$$

Since F is continuous everywhere except at $x = 0$, we must show that at all points where $x \neq 0$,

$$\lim_{n\to\infty} F_n(x) = F(x),$$

where, for $n \geq 1$, F_n is the distribution function of X_n. Clearly, for $x < 0$, $F_n(x) = 0$, and $\lim_{n\to\infty} F_n(x) = 0 = F(x)$. Similarly, for $x \geq 1$, $F_n(x) = 1$ and $\lim_{n\to\infty} F_n(x) = 1 = F(x)$. For $0 < x < 1$, it follows that

$$\begin{aligned} F_n(x) &= P(X_n \leq x) = P\big(\min(U_1, U_2, \ldots, U_n) \leq x\big) \\ &= 1 - P\big(\min(U_1, U_2, \ldots, U_n) > x\big) = 1 - P(U_1 > x, U_2 > x, \ldots, U_n > x) \\ &= 1 - P(U_1 > x)P(U_2 > x) \cdots P(U_n > x) = 1 - \Big(\frac{1-x}{1-0}\Big)^n = 1 - (1-x)^n. \end{aligned}$$

Clearly,

$$\lim_{n\to\infty} F_n(x) = \begin{cases} 0 & x < 0 \\ \\ 1 & x > 0 \end{cases}$$

This shows that except at $x = 0$, $\lim_{n\to\infty} F_n(x) = F(x)$. ♦

11B PROPORTION VERSES DIFFERENCE IN COIN TOSSING

Suppose that a fair coin is flipped independently and successively. Let $n(H)$ be the number of times that heads occurs in the first n flips of the coin. Let $n(T)$ be the number of times that tails occurs in the first n flips. By the strong law of large numbers, with probability 1,

$$\lim_{n\to\infty} \frac{n(H)}{n} = \lim_{n\to\infty} \frac{n(T)}{n} = \frac{1}{2}.$$

That is, with probability 1, the *proportion* of heads and the *proportion* of tails, as $n \to \infty$, is 1/2. This result is subject to misinterpretation. Some might think that if a fair coin is tossed a very large number of times, then heads occurs as often as tails. For example, Bertrand Russell (1872–1970), in his *Religion and Science* (Oxford University Press, 1947, page 158), writes

> It is said (though I have never seen any good experimental evidence) that if you toss a penny a great many times, it will come heads about as often as tails. It is further said that this is not certain, but only extremely probable.

None of these statements is false. But they are subject to interpretations.

In Chapter 12, when discussing recurrent states of symmetric simple random walks, we have shown that, in successive and independent flips of a fair coin, it happens that at times the number of heads obtained is equal to the number of tails obtained. In that chapter, we will show that, with probability 1, this happens infinitely often, but the expected number of tosses between two such consecutive times is ∞. Hence, for some large number of tosses, what Bertrand Russell quotes is accurate. However, in general, when we say heads occurs "about as often" as tails, it is important to note that the fuzzy term about as often is a relative term and should be interpreted only relative to the number of times the coin is flipped. Frequently, when a fair coin is flipped a large number of times, in the usual sense, one side will not come "about as often" as the other side. But relatively speaking, it does. For example, we will show below that if a fair coin is flipped independently and successively 10^{20} times, then the probability exceeds 0.96 that the absolute value of the difference between the number of heads and the number of tails obtained is at least half a billion. This is a very large difference. Nevertheless when compared with the huge number of tosses, 10^{20}, it is insignificant. In fact, it is straightforward to show that in 10^{20} tosses of the coin, almost certainly, the difference between the number of heads and the number of tails is less than 40 billion. Considering the fact that 10^{20} is one hundred billion billion, relative to 10^{20}, 40 billion is really a small number. It is $\dfrac{1}{25 \times 10^9}$th of the number of times the coin is flipped. Therefore, relatively speaking, when the coin is flipped 10^{20} times, still "it will come heads about as often as tails."

We will end this section by showing that in 10^{20} flips of a fair coin, successively and independently, the probability exceeds 0.96 that the absolute value of the difference between the number of heads and the number of tails obtained is at least half a billion. Let X be the number of heads obtained. Clearly, X is a binomial random variable with parameters 10^{20} and 1/2. Therefore,

$$E(X) = \frac{1}{2} \cdot 10^{20} \quad \text{and} \quad \sigma_X = \sqrt{10^{20}\Big(\frac{1}{2}\Big)\Big(1 - \frac{1}{2}\Big)} = \frac{1}{2} \cdot 10^{10}$$

To show that

$$P\big(\big|X-(10^{20}-X)\big| \geq \frac{1}{2}\cdot 10^9\big) \geq 0.96,$$

we will use the De Moivre-Laplace theorem, and we ignore the correction for continuity since its effect, when dealing with large numbers, is negligible. We have

$$\begin{aligned}
P\Big(\big|X-(10^{20}-X)\big| &\geq \frac{1}{2}\cdot 10^9\Big) \\
&= P\Big(\big|2X-10^{20}\big| \geq \frac{1}{2}\cdot 10^9\Big) \\
&= P\Big(2X-10^{20} \geq \frac{1}{2}\cdot 10^9 \quad \text{or} \quad 2X-10^{20} \leq -\frac{1}{2}\cdot 10^9\Big) \\
&= P\Big(X \geq \frac{1}{2}\cdot 10^{20} + \frac{1}{4}\cdot 10^9\Big) + P\Big(X \leq \frac{1}{2}\cdot 10^{20} - \frac{1}{4}\cdot 10^9\Big) \\
&= P\left(\frac{X-\frac{1}{2}\cdot 10^{20}}{\frac{1}{2}\cdot 10^{10}} \geq 0.05\right) + P\left(\frac{X-\frac{1}{2}\cdot 10^{20}}{\frac{1}{2}\cdot 10^{10}} \leq -0.05\right) \\
&\approx \int_{0.05}^{\infty} \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2}\,dx + \int_{-\infty}^{-0.05} \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2}\,dx \\
&= 1-\Phi(0.05)+\Phi(-0.05) = 2\big[1-\Phi(0.05)\big] \\
&\approx 2(1-0.5199) = 0.9602.
\end{aligned}$$